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TECHNICAL SUMMARY

PV-STATCOM

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Undergraduate thesis · IOE Pulchowk · 2018

ABSTRACT

A dual-function inverter that injects solar power by day and gives reactive power support when there is little or no PV generation. This project developed a dual-function inverter control. Under irradiance the inverter works as a normal grid-connected PV inverter, and during low or no PV generation it works as a STATCOM for reactive power compensation, making fuller use of hardware that is already installed. This was Kishor's bachelor's thesis and was published in the Journal of the Institute of Engineering.

1. Problem and motivation

A grid-connected PV inverter is fully used only around midday and sits idle at night. Meanwhile the grid still needs reactive power support to hold voltage, which usually means buying dedicated STATCOM hardware.

2. Methodology

- Developed dual-function control: grid-connected PV inverter under irradiance, STATCOM during low or no PV generation
- Designed switching and control logic to move dynamically between active power injection and reactive power support
- Implemented and validated the control in MATLAB/Simulink for varying irradiance and grid conditions

3. Figures

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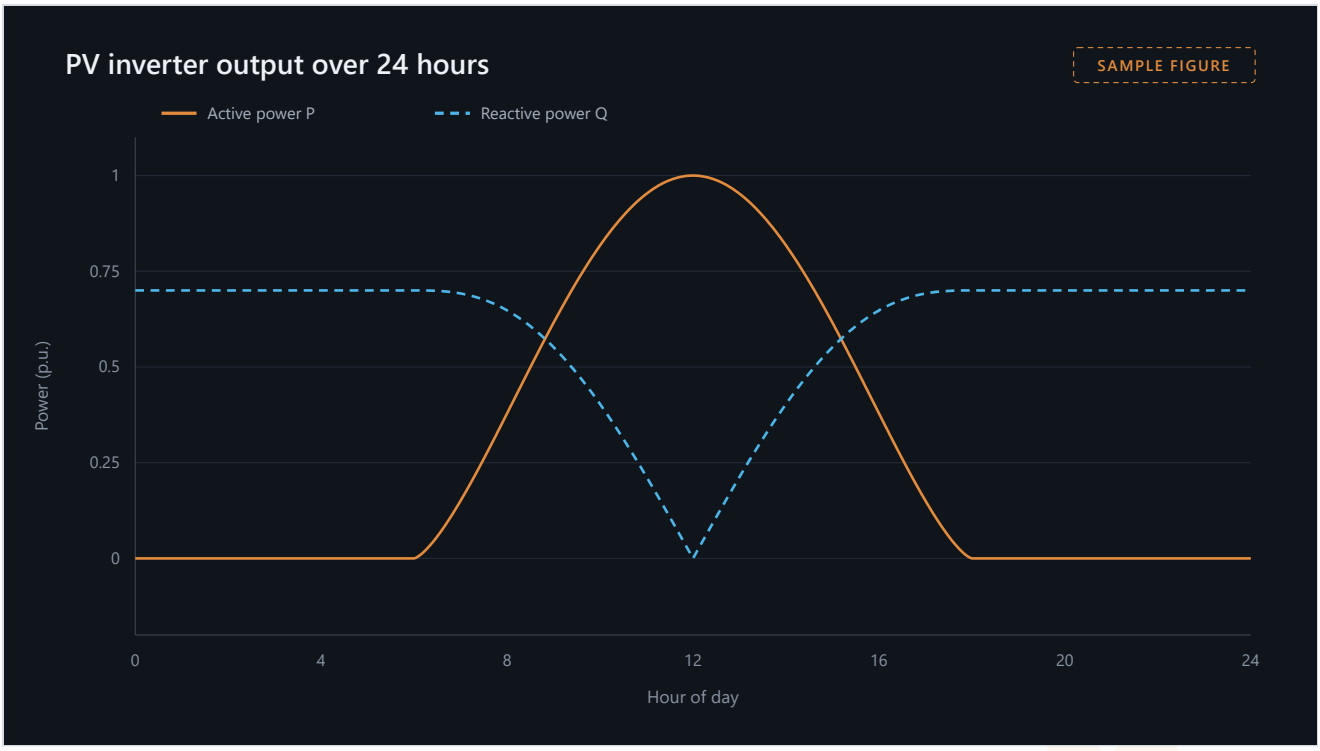


Figure 1. Active power by day, reactive power support at night.

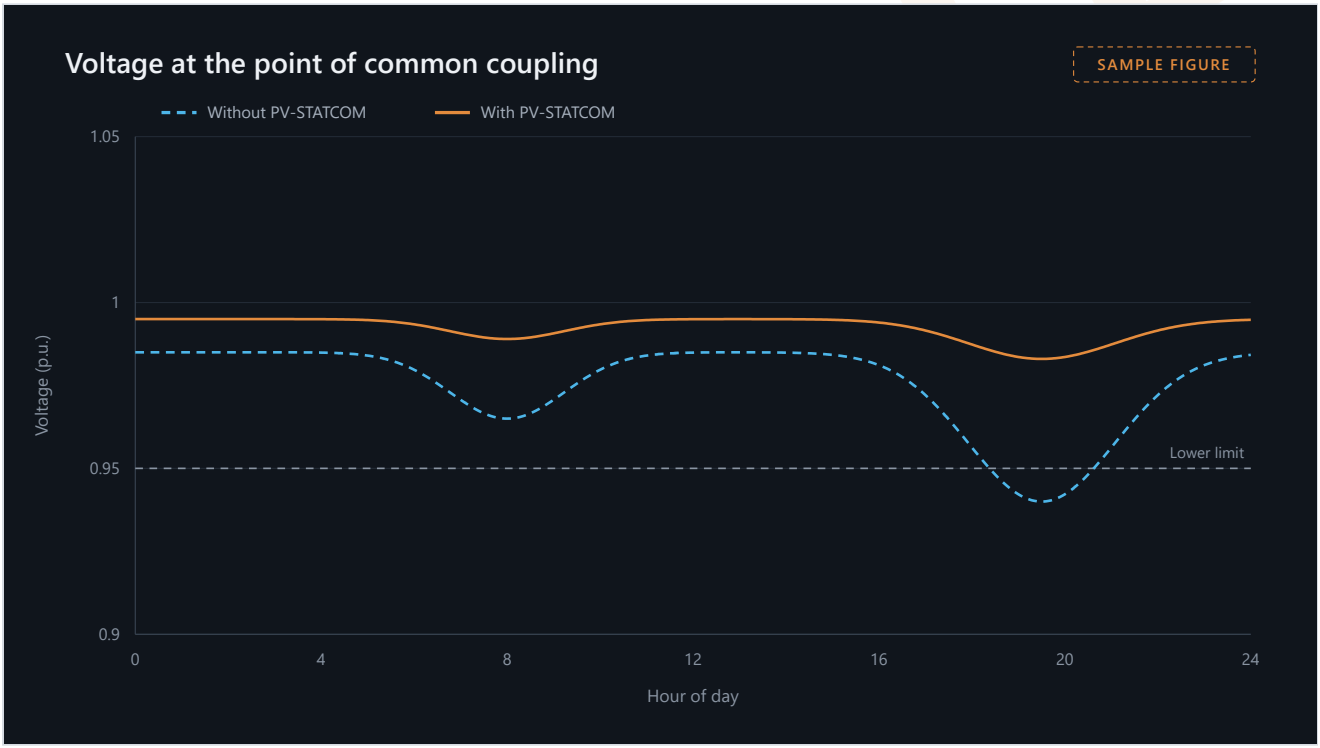


Figure 2. Voltage at the point of common coupling with and without PV-STATCOM.

4. Results

- Published in the Journal of the Institute of Engineering, vol. 15, no. 3, pp. 411–419 (2019)

5. Team and contribution

Contribution: Thesis project and co-author. Designed the switching and control logic and built the MATLAB/Simulink models.

Tools: MATLAB/Simulink, PSIM, PID control

6. References

1. Paper (DOI): https://doi.org/10.3126/jie.v15i3.32762

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